

Temporal Anchors and the Recursive Packet-Transport Wall

A Typed Source Repair and Outward Radius Audit

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Abstract

RH-149 proposed propagating the ten RH-142 spectral-packet balls through the RH-143 threshold branches of the RH-96 recursive refresh. We show first that this composition is not type-correct. RH-142 certifies a rank-four packet of the postblock state $A^{H(\sigma)}S$, whereas RH-96 starts at time zero from the source S and uses clock ranks 4, 5, 6, 6, 7. Orthogonal projectors of different ranks have operator distance one. Eight channels therefore fail by rank alone; on the two rank-four channels, the temporal center distances are 0.8652 and 0.9963, and adding the inherited radii 0.6097 and 0.9574 makes both transferred balls vacuous. None of the ten RH-142 balls is an informative RH-96 seed.

We repair both types. At time zero we use the native clock rank and enclose the exact-binary moving-memory Grams around their numerical centers in 384-bit Arb arithmetic. All ten source packets certify; the smallest source gap is 3.303×10^{-4} and the largest projector radius is 2.876×10^{-13} . We then prove an outward one-step recursion through the projected cross, clipped threshold branch, selected left-singular subspace, enriched projector, and Ritz packet. The direction gate correctly uses $s_4 - s_5$ when the clock rank exceeds four, even though maximum clipping makes s_5 irrelevant to the branch count.

The repaired finite audit still closes no complete chain. Seventeen updates certify in total, but every channel stops at update two or three: seven first at the branch gate and three first at the Ritz gate. The smallest decisive failure ratio is 1.506. This is a universal-information obstruction, not a disproof of the exact recursion. It redirects the next layer toward a joint, gauge-free output-packet enclosure that avoids separately resolving weak enrichment directions.

1 Packet balls carry time and rank types

Let $P_{t,r}$ denote a rank- r orthogonal packet projector attached to time t . A ball around $P_{t,r}$ is not an untyped approximation to every packet appearing elsewhere in a construction. Both labels matter before any perturbation theorem can be composed.

Theorem 1 (Typed anchor compatibility). *Let P, Q be finite-dimensional orthogonal projectors.*

1. If $\text{rank } P \neq \text{rank } Q$, then $\|P - Q\| = 1$.
2. If the ranks agree and $\|\tilde{P} - P\| \leq \rho$, then

$$\|\tilde{P} - Q\| \leq \min\{1, \|P - Q\| + \rho\}.$$

Consequently an inherited radius is informative at the target anchor only when the ranks agree and the transferred upper is below one.

Proof. If $\text{rank } P > \text{rank } Q$, dimension counting gives a unit vector $v \in \text{ran } P \cap \ker Q$. Hence $(P - Q)v = v$, while the difference of two orthogonal projectors has norm at most one. Interchanging P, Q handles the other rank ordering. The second claim is the triangle inequality together with the unit diameter of the projector metric. $\square$

The theorem immediately exposes the proposed interface error. RH-142 uses rank four at time $H(\sigma)$. RH-96 uses ranks 4, 5, 6, 6, 7 at time zero. Thus eight transfers are exactly unit radius. For $\sigma = 0.16$, where the ranks agree, the two center-distance-plus-radius sums are 1.4749 and 1.9537. Capping at one again leaves no information.

Merely moving the RH-142 rank-four truncation to time zero is also not a uniform repair. If p_i are normalized squared source singular values, $q = \sum_{i>4} p_i$, and $c = 1 - q$, the exact ideal-truncation formulas are

$$g_4 = \frac{p_4}{c}, \quad e_4 = \max\left\{p_5, \frac{qp_1}{c}\right\}.$$

Only three of ten frozen source spectra satisfy $g_4 > 2e_4$. This diagnostic assumes an ideal SVD and is not itself an interval certificate, but it shows why the native clock rank must be retained.

2 Source-aligned repair

Write

$$G_0 = \frac{S^* S}{\|S\|_F^2}, \quad G_t = \frac{S_t^* S_t}{\|S_t\|_F^2} + \eta G_{t-1}, \quad S_t = A^t S, \quad \eta = 2^{-9}.$$

The exact model treats every archived fp64 entry of A, S as its exact binary value. At each time the audit evaluates the normalized snapshot and memory recurrence in 384-bit Arb arithmetic, then encloses it around the archived fp64 Gram B_t . We use the Frobenius upper as an operator radius δ_t .

For the source center B_0 , a symmetric eigendecomposition supplies the clock-rank gap; its reconstruction error and a conservative local backward proxy are subtracted twice. If the resulting gap g_0 obeys $g_0 > 2\delta_0$, Davis–Kahan gives [Davis and Kahan, 1970, Bhatia, 2007]

$$\epsilon_0 \leq \frac{\delta_0}{g_0 - \delta_0}.$$

All ten source packets pass. The minimum g_0 is 3.303289×10^{-4} , the maximum memory-Gram radius over every audited time is 2.642×10^{-15} , and $\max \epsilon_0 = 2.876 \times 10^{-13}$. The corrected seed is therefore extremely tight; the subsequent loss does not originate at the source.

3 Outward recursive transport

For a Hermitian Gram B and rank- r packet Q , define the projected cross

$$\mathcal{K}(B, Q) = (I - Q)BQ.$$

Suppose exact objects $\tilde{B}, \tilde{Q}$ satisfy

$$\|\tilde{B} - B\| \leq \delta, \text{quad}\|\tilde{Q} - Q\| \leq \epsilon, \text{quad}\|B\| \leq b.$$

The first outward radius is

$$\eta_K = \delta + 2b\epsilon, \text{quad}\|\mathcal{K}(\tilde{B}, \tilde{Q}) - \mathcal{K}(B, Q)\| \leq \eta_K. \quad (1)$$

Indeed, split the matrix perturbation from the projector perturbation and use

$$\mathcal{K}(B, P) - \mathcal{K}(B, Q) = (Q - P)BP + (I - Q)B(P - Q).$$

Let $s_1 \geq s_2 \geq \dots$ be the nominal cross singular values. The RH-96 policy clips the relative threshold $\tau = 10^{-8}$ to widths $2 \leq w \leq 4$. RH-143 gives the exact open branch radius

$$\beta = \min \left\{ \frac{s_w - \tau s_1}{1 + \tau} \ (w > 2), \quad \frac{\tau s_1 - s_{w+1}}{1 + \tau} \ (w < 4) \right\},$$

where irrelevant clipped sides are omitted. The branch is fixed when $\eta_K < \beta$.

Branch stability does not identify the selected direction space. Put

$$\gamma = s_w - s_{w+1},$$

using the actual next singular value. In particular, when $w = 4$ and $r > 4$, one must use s_5 rather than zero. Wedin's theorem gives, under $\gamma > 2\eta_K$ [Wedin, 1972, Stewart and Sun, 1990],

$$\|\tilde{D} - D\| \leq \epsilon_D := \frac{\eta_K}{\gamma - \eta_K}. \quad (2)$$

Here $D, \tilde{D}$ are the selected left-singular projectors. Since each direction space is orthogonal to its input packet, the enriched projectors $E = Q + D$ obey

$$\|\tilde{E} - E\| \leq \epsilon_E := \min\{1, \epsilon + \epsilon_D\}.$$

Finally set $H = EBE$ and $\tilde{H} = \tilde{E}\tilde{B}\tilde{E}$. Then

$$\|\tilde{H} - H\| \leq \zeta := \delta + 2b\epsilon_E. \quad (3)$$

If the nominal rank- r Ritz gap g_R satisfies $g_R > 2\zeta$, the output packet radius is

$$\epsilon^+ = \frac{\zeta}{g_R - \zeta}.$$

σ	side	rank	source radius	certified updates	first wall	ratio
0.16	L	4	5.36×10^{-15}	1	branch	1.52×10^8
0.16	R	4	2.65×10^{-15}	1	branch	3.94×10^8
0.08	L	5	3.77×10^{-15}	1	Ritz	1.23×10^2
0.08	R	5	6.88×10^{-15}	2	branch	4.62×10^6
0.04	L	6	2.90×10^{-14}	2	branch	48.4
0.04	R	6	5.30×10^{-15}	2	branch	2.98
0.02	L	6	9.66×10^{-14}	2	branch	1.51
0.02	R	6	1.29×10^{-14}	2	Ritz	32.9
0.01	L	7	2.88×10^{-13}	2	branch	5.11
0.01	R	7	4.66×10^{-14}	2	Ritz	4.27

Table 1: Source-aligned outward transport at the primary threshold.

Theorem 2 (One-step packet transport). *If the branch, direction, and Ritz inequalities above are all strict, then the exact update takes the same clipped width as the nominal update and its rank- r output projector lies in the radius- ϵ^+ ball around the nominal Ritz packet.*

Proof. Equation (1) and singular-value Weyl bounds place every exact cross singular value in its nominal η_K interval. The sharp RH-143 branch radius fixes the clipped count. Wedin’s theorem gives (2); orthogonality gives the enriched-projector sum; and expanding $\tilde{E}\tilde{B}\tilde{E} - EBE$ gives (3). Davis–Kahan at the Ritz gap completes the update. $\square$

The strict gates are universal-information barriers. At a threshold contact the clipped branch can change. If a singular or Hermitian gap is at most twice its perturbation radius, diagonal two-mode examples permit the selected and omitted values to meet or reverse. A particular correlated exact matrix may still behave well, but norm-ball data alone cannot identify which side occurs.

4 Finite obstruction audit

The audit applies Theorem 2 sequentially and stops at the first failed gate; it never substitutes a nominal branch after certification is lost. Local fp64 cross and Ritz backward proxies are charged in addition to the Arb Gram radii. Table 1 records the result. “Ratio” means η_K/β at a branch wall or $2\zeta/g_R$ at a Ritz wall, so a failure has ratio at least one.

There are 17 certified recursive updates, but no complete chain. Seven channels first fail the branch gate; all seven simultaneously fail the rank-aware direction gate. Three first fail only after enrichment, at the Ritz gap. The nearest decisive wall is the $\sigma = 0.02$ left branch with ratio 1.5056; the others range to 3.94×10^8 .

The broad radii are generated after one or two updates even though the source radii are tiny. The separated construction first resolves a weak direction subspace and only then asks whether its enriched Ritz packet is stable. When $s_4 - s_5$ is small, direction uncertainty is amplified before the Ritz step; the next cross can then exceed even a positive nominal branch margin.

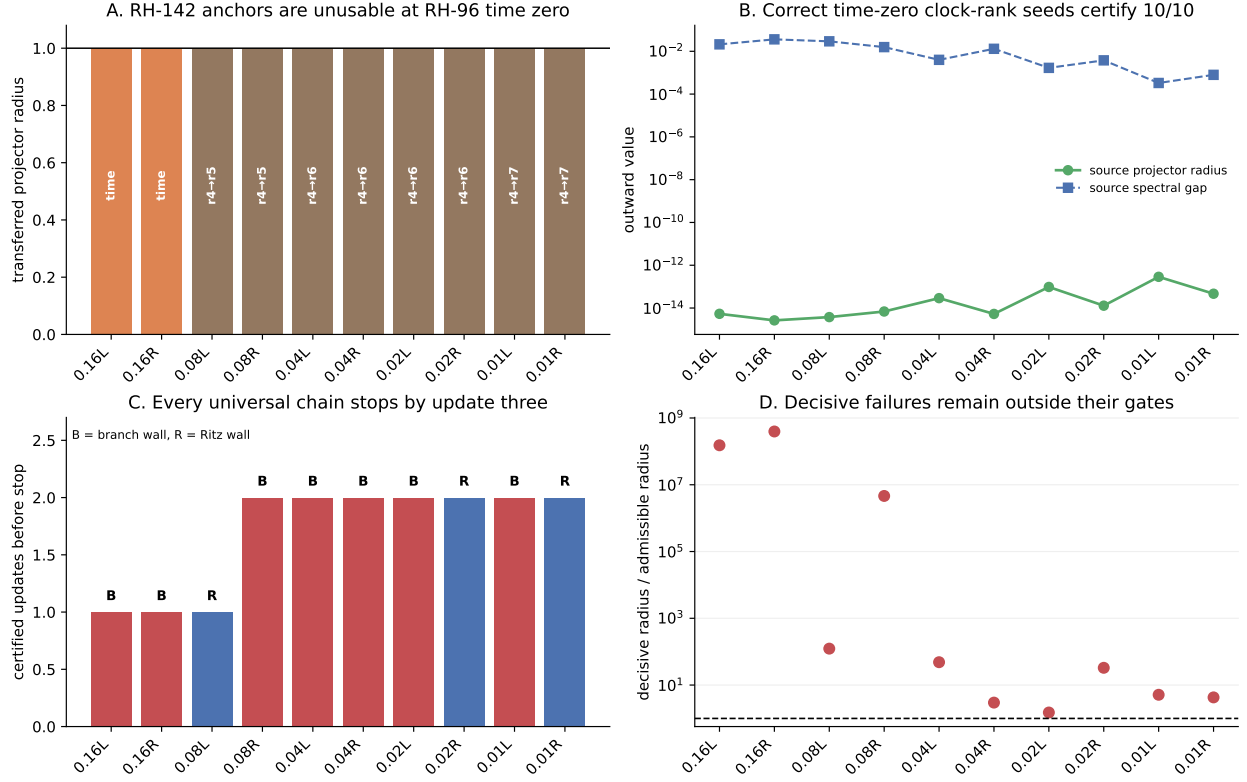


Figure 1: Inherited type obstruction, corrected source packets, certified prefix lengths, and decisive failure ratios.

5 Route consequence and claim boundary

The original RH-149 instruction “propagate the RH-142 packet balls through RH-143” cannot be executed: those balls have the wrong temporal or rank type. Correcting the seed removes that problem completely but exposes the next one. The modular norm-ball route loses correlation between the input packet, cross, selected weak directions, and output Ritz packet.

The next layer should therefore enclose the output packet jointly, for example through a graph transform, block Riccati equation, residual theorem, or contour projector. Such a result may bypass unstable coordinates inside a weak direction cluster. A fallback is to define a new recursion whose native start is the postblock horizon certified by RH-142. Transported outward assembly must wait until one of these routes closes a complete packet chain.

We have proved a typed-anchor theorem, a one-step outward transport theorem, ten corrected finite source certificates, and a finite universal-information obstruction. We have not disproved the exact RH-96 chain, closed the E_{update} interface, built the RH-138 outward recurrence on interval packets, proved any all-level packet law or Stage A statement, constructed a Hilbert–Polya operator, identified zeta zeros, or proved the Riemann Hypothesis.

References

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